

Abstract

Gravitational waves have intrinsic uncertainties between the true waveform we detect from space and the waveform models, or waveform approximants, we use for parameter estimation. These uncertainties can be quantified in amplitude and phase, δA and $\delta \phi$. We want to examine the effect they have on the posterior data of the parameter estimation: how much does waveform uncertainty matter? This can be achieved by marginalizing over a large set of potential waveform uncertainties after the parameter estimation process. To do so, we need an efficient method of generating waveform uncertainty data and accessing it. This process is typically long and inefficient, as multiple generated waveforms are needed for each possible set of source parameters. For example, for a GW170817 posterior this takes between 10 and 20 minutes to complete. We wanted a way to store this uncertainty data in a file, where it can be easily accessed to greatly reduce the amount of time it takes to generate a dataset for our needs. The solution was to parameterize the waveform uncertainty. Rather than expressing each draw of waveform uncertainty as a fixed set of hundreds of thousands of points, each one was expressed as 15 parameters. These parameters were then used to interpolate the waveform uncertainty over any number of frequency points in a matter of seconds, improving the efficiency of generating the waveform uncertainty data to be marginalized over.

Calculating the Uncertainties

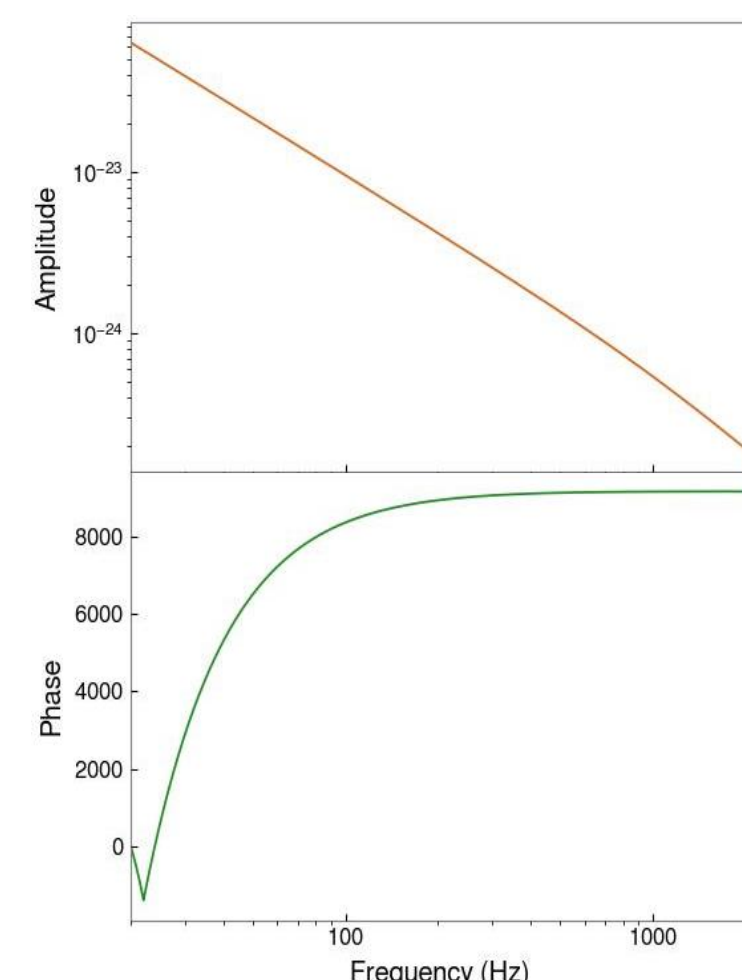


Figure 1 displays 100 frequency domain waveforms based on GW170817 source parameters expressed as their amplitude and phase.

We take the differences between the waveforms of two different approximants to be adequate representations of waveform uncertainty. We choose the models:

IMRPhenomPv2_NRTidalv2
SEOBNRv4T_surrogate

The amplitude uncertainty, δA , and residual phase uncertainty, $\delta \phi_R$, are calculated from the waveforms with the following expressions:

$$S_{IMR} = \{h_1, h_2, h_3, \dots, h_n\}$$

$$S_{EOB} = \{g_1, g_2, g_3, \dots, g_n\}$$

$$f = \{f_1, f_2, f_3, \dots, f_n\}$$

$$h_n, g_n \in \mathbb{C}$$

$$\delta A_n = \frac{|h_n| - |g_n|}{|h_n|}$$

$$\delta \phi_n = \tan^{-1} \frac{\text{Im}[h_n]}{\text{Re}[h_n]} - \tan^{-1} \frac{\text{Im}[g_n]}{\text{Re}[g_n]}$$

$$\delta A = \{\delta A_1, \delta A_2, \delta A_3, \dots, \delta A_n\}$$

$$\delta \phi = \{\delta \phi_1, \delta \phi_2, \delta \phi_3, \dots, \delta \phi_n\}$$

$\delta \phi$ can drift between the models from an overall shift in time and phase density. To remove this effect, we take a regression line weighted by the power spectral densities¹. Subtracting this line from $\delta \phi$ removes the best fit overall shifts, ϕ_0 and t_0 , resulting in residual phase uncertainty.

$$\delta \phi_R = \delta \phi - \phi_0 - 2\pi f t_0$$

In practice, disagreements between the approximants on how tidal effects end the waveform can lead to discontinuities in δA , as shown by the following limit:

$$\lim_{|h_n| \rightarrow 0^+} \delta A_n = \lim_{|h_n| \rightarrow 0^+} \frac{|h_n| - |g_n|}{|h_n|} = -\infty$$

Corrections to the waveform uncertainties are needed to remove the discontinuities.

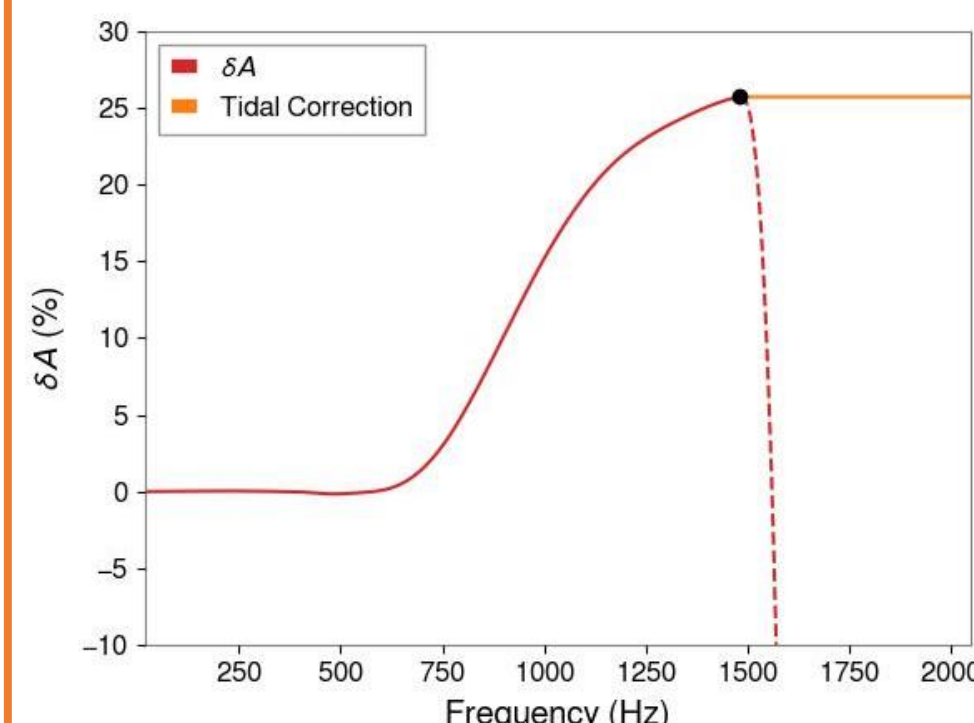


Figure 2. Example of the tidal correction in δA .

We hold the uncertainties constant starting right before the discontinuity to the end of the frequency grid. This is done to ensure that every frequency point has an associated δA value.

$\delta \phi_R$ also shows a discontinuity at the same frequency as its δA counterpart, so it is treated with the same correction.

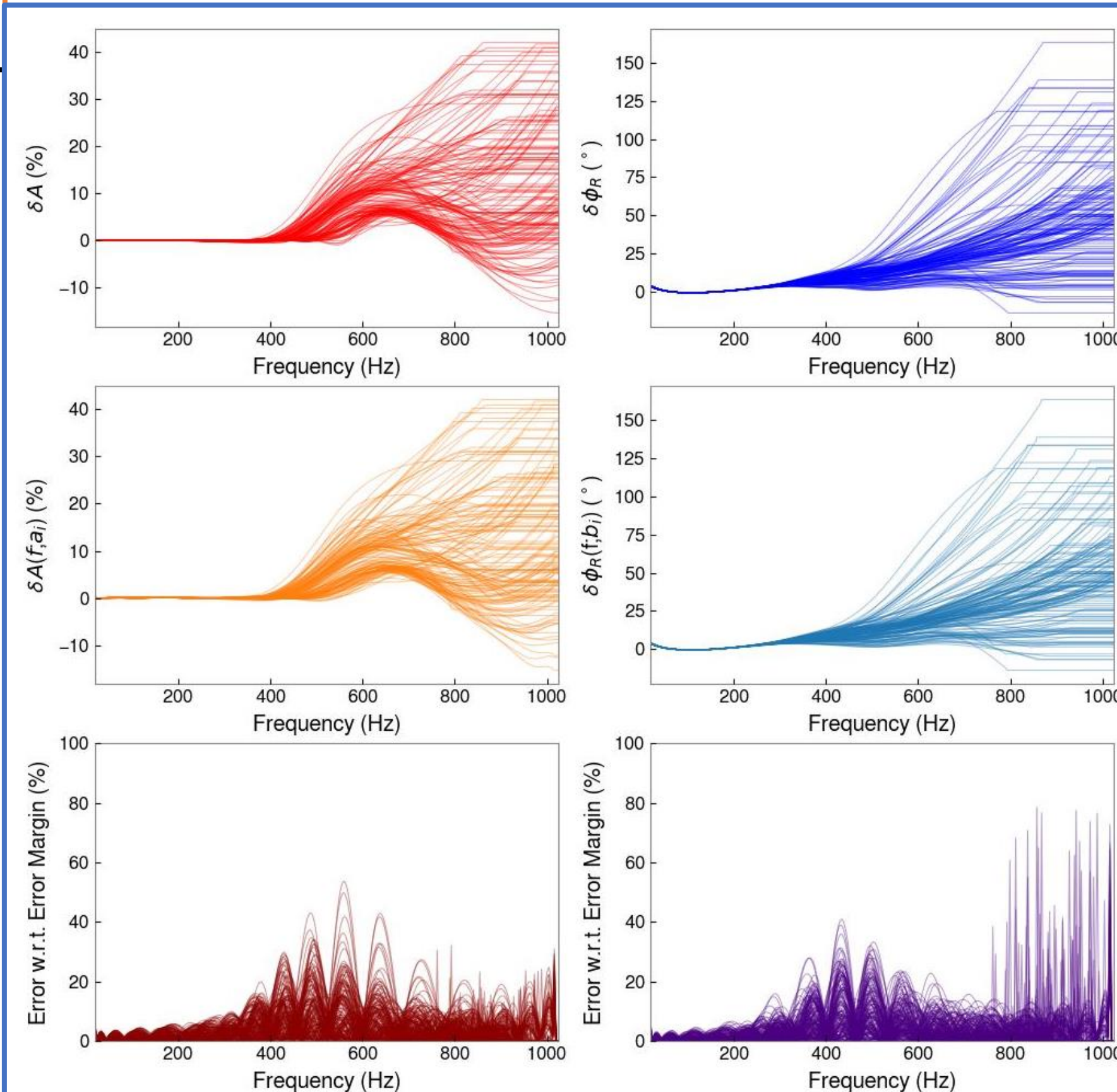


Figure 3. 150 draws of waveform uncertainties, their parameterizations with $N=15$, and the errors between them. The error is measured as the difference between the uncertainty and its parameterization relative to the error margin defined below. Here, 100% error means that the error exceeded the margin, and the parameterization is not valid.

Parameterization

The sets of waveform uncertainty and frequency are all the same length, but the arrays are large in scale:

$$\text{len}|\delta A| = \text{len}|\delta \phi_R| = \text{len}|f| > 2.1 \times 10^5$$

When dealing with thousands of sets of waveform uncertainty, this size is too big to be dealt with efficiently. To reduce the size of each data set, we look to fit a function of frequency to each waveform uncertainty. Before we did this, we needed to find how precise these fits needed to be. We defined the following error margins from calibration uncertainty³:

2% Uncertainty in Amplitude
2° Uncertainty in Phase

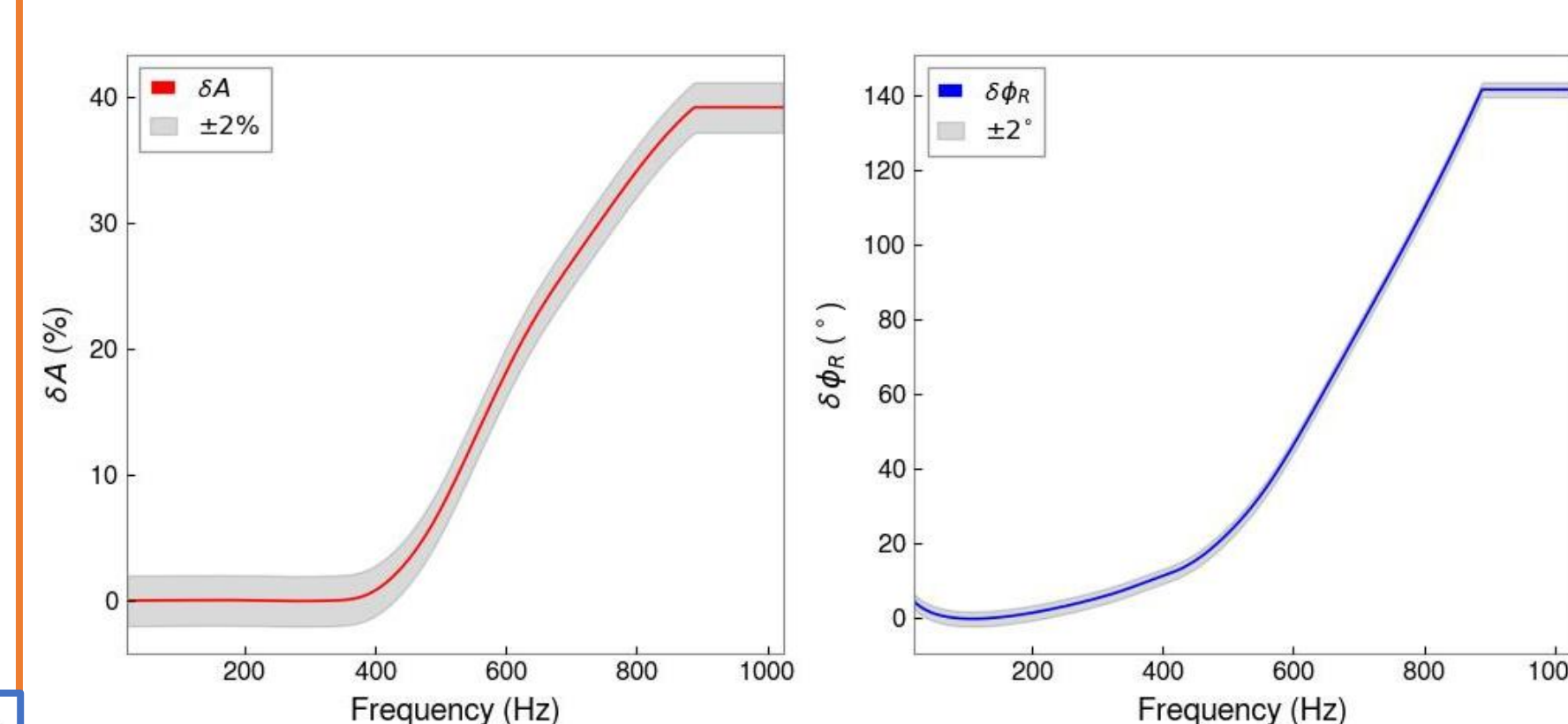


Figure 4. Error margins (in gray) shown around an example set of waveform uncertainty. Parameterizations of waveform uncertainty are considered valid if they fall inside this region.

With the error margin defined, we fit a series of Chebyshev Polynomials with N terms to the uncertainties:

$$T_0(x) = 1$$

$$T_1(x) = x$$

$$T_2(x) = 2x^2 - 1$$

$$T_3(x) = 4x^3 - 3x$$

$$T_4(x) = 8x^4 - 8x^2 + 1$$

$$T_5(x) = 16x^5 - 20x^3 + 5x$$

$$T_6(x) = 32x^6 - 48x^4 + 18x^2 - 1$$

$$T_7(x) = 64x^7 - 112x^5 + 56x^3 - 7x$$

$$T_8(x) = 128x^8 - 256x^6 + 160x^4 - 32x^2 + 1$$

$$T_9(x) = 256x^9 - 576x^7 + 432x^5 - 120x^3 + 9x$$

$$\delta A(f; a_i) = \sum_{i=0}^{N-1} a_i T_i(f) \quad \delta \phi_R(f; b_i) = \sum_{i=0}^{N-1} b_i T_i(f)$$

$\{a_0, \dots, a_{N-1}\}$ and $\{b_0, \dots, b_{N-1}\}$ would be the sets of parameters that represent the waveform uncertainties, where N is the number of parameters wanted.

Figure 5. The first 10 Chebyshev polynomials². The curve-fits are expressed as a series of these polynomials, each multiplied by a constant, a_i and b_i . These N constants became the parameters that defined the parameterized waveform uncertainties.

Conclusions and Future Work

Using this method of parameterization, we reduced the size of a set of waveform uncertainty from over 210,000 points to just 15 coefficients. This reduces the amount of data by over 99.99%! Large sets of these parameterized waveforms are being used in multiple projects:

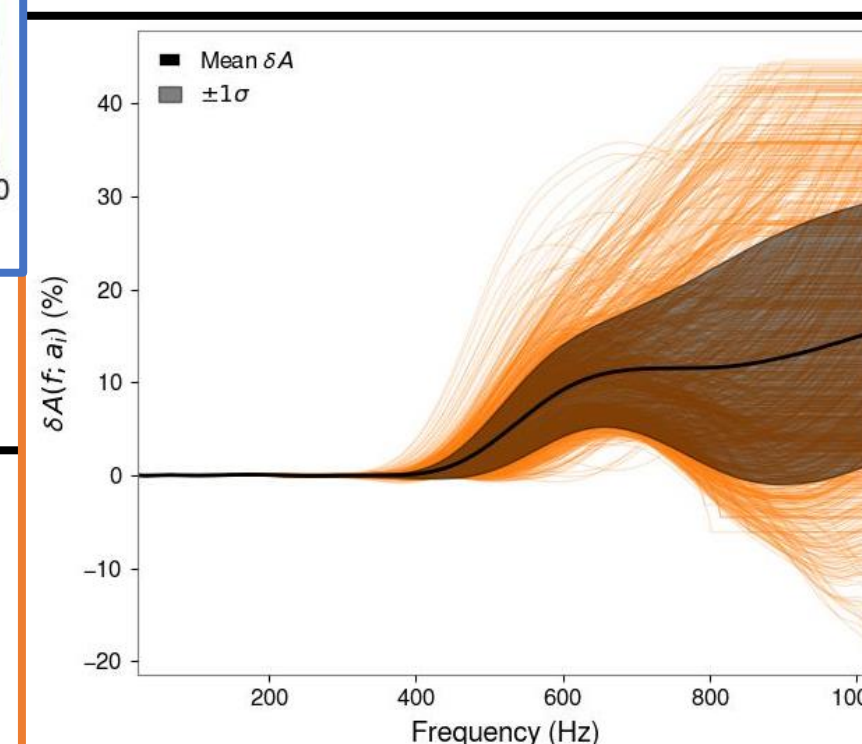


Figure 6. Taking the mean and standard deviation of a large collection of waveform uncertainty to be used in marginalization. This will reweight a PE posterior to correct for waveform uncertainty.

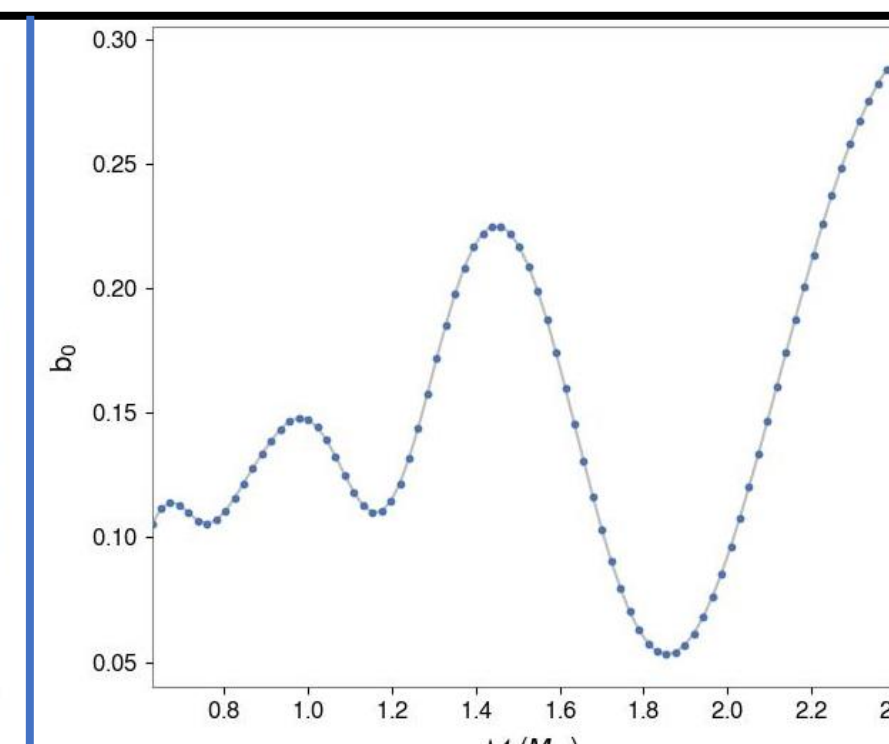


Figure 7. Using the parameterized waveform uncertainty to look for a relationship between waveform uncertainty and BNS source parameters. Here we look at the first coefficient for the $\delta \phi_R$ parameterization and chirp mass.

References

- ¹LIGO Document P1900011-v1, "Power Spectral Densities (PSD) release for GWTC-1"
- ²Z. Omar, N. Mitianoudis, T. Stathaki, "Two-dimensional Chebyshev Polynomials for Image Fusion" (2010).
- ³L. Sun, E. Goetz, J. S. Kissel, J. Betzwieser, S. Karki, A. Viets, M. Wade, D. Bhattacharjee, V. Bossilkov, P. B. Covas, L. E. H. Datrier, R. Gray, S. Kandhasamy, Y. K. Leconte, G. Mendell, T. Mistry, E. Payne, R. L. Savage, A. J. Weinstein, S. Aston, A. Buikema, C. Cahillane, J. C. Driggers, S. E. Dwyer, R. Kumar, and A. Urban, "Characterization of systematic error in Advanced LIGO calibration" (2020).